Poster: Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction

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CCS CONCEPTS

• **Human-centered computing → Ubiquitous and mobile computing**; • **Computing methodologies → Learning from implicit feedback**.

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1 INTRODUCTION

Addiction is a brain disease manifested by the loss of control over drugs or behaviors, despite negative consequences. Although addiction research has been conducted for decades in psychiatry and neuroscience, a comprehensive understanding of the mechanisms underlying addiction has not yet been achieved. Recent studies in neuroscience [1] have sought to bring light upon this issue by measuring exploration/exploitation decisions in sequential choice tasks, requiring balancing the need to exploit known options and to explore new ones. These studies show a relationship between addiction and exploration/exploitation decisions. For example, people addicted to substances (e.g. alcohol or methamphetamine) or behaviors (e.g. gambling) have tendencies to explore less, which implies they have difficulties "seeing the big picture".

There is a small yet growing literature modeling explore/exploit decisions of addicted people through inverse reinforcement learning (IRL) [4]. In this previous work, the models made by decision history of addicted people have higher learning weights and probability to exploit than those of normal people, which means that addicted people are more sensitive to their most recent activity. However, existing methods to measure exploration/exploitation decisions are lab-based game experiments such as n-armed bandit [3] or clock task [6], which are high cost, time-consuming and not scalable. Therefore, they are not suitable for modeling through IRL, which requires large behavioral trajectories. In this work, we argue for the first time that mobile sensing is a more cost-efficient and scalable alternative to the lab-based game experiments for understanding and modeling the mechanisms of addiction (Figure 1).

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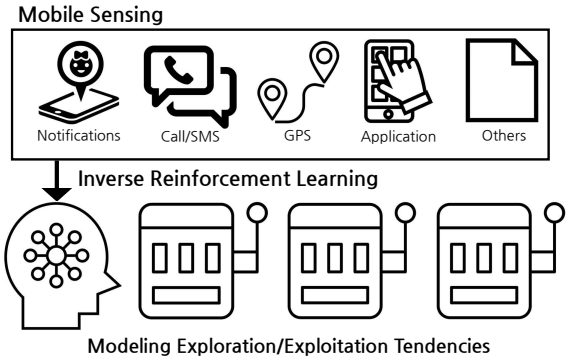

Figure 1: Inverse reinforcement learning (IRL) can model exploration/exploitation decisions.

2 APPROACH

We plan to show mobile sensing can measure people's exploration and exploitation tendencies in a more accurate and scalable way than the existing lab-based game experiments. For this, we plan to run a within-subjects experiment where participants test both conditions: (i) solving a n-armed bandit task and (ii) collecting their behavior through mobile sensing. For ground truth, the participants will be divided into addicted and normal groups based on self-reported rating scale for addiction. In this experiment, we will model a reward function, which approximates brain mechanisms, with behavior trajectories in each condition through IRL. We will not only compare the accuracy of the resulting models, but also the interpretability. These models can be represented with a Markov decision process, so graph visualization techniques can be adopted. From this, we will implement a visual analytic system for psychiatrists to understand addiction through visual data mining.

We plan to collect notification, call, SMS, GPS and application logs through mobile sensing. We expect these features mirror a user's exploration/exploitation tendency. Previously, Lee's work [5] showed the correlation, the faster the response to an alarm, the greater the risk of smartphone overuse. This result is consistent with other addiction studies from psychiatry and neuroscience, which show that addiction makes people become more sensitive to recent rewards, and unable to consider long term consequences of decisions. Also, we expect that social and geo-spatial exploration, measured by call, SMS and GPS history, can correlate with exploratory and exploitative preferences, since social and geo-spatial exploration have been considered as novelty seeking behaviors in mobile sensing research [2]. We ultimately aim to foster closer collaboration between mobile-sensing specialists and psychiatrists.

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